PORTFOLIO CASE STUDY

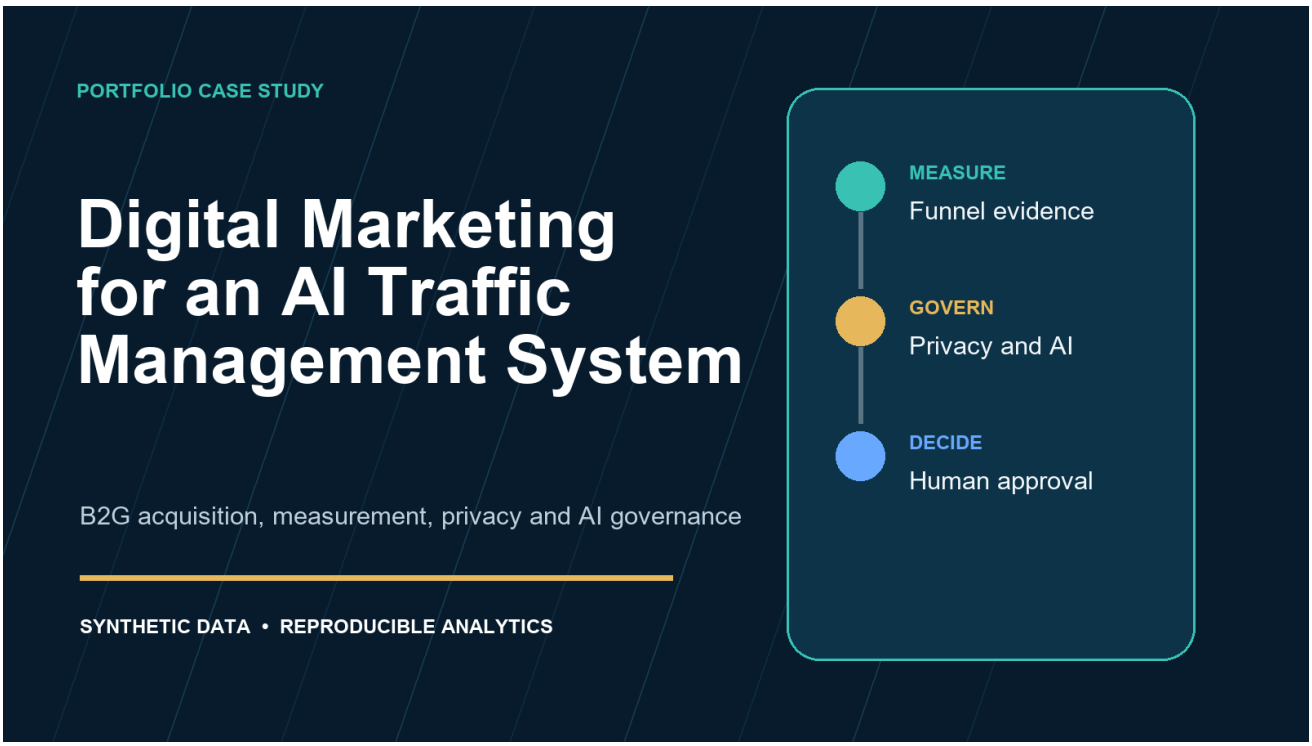
Digital Marketing for an AI Traffic Management System

An evidence-led B2G acquisition and measurement design for municipal traffic-management pilots

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Scope notice. This is an educational portfolio scenario. All campaign performance is synthetic. The report does not describe a real company, customer, deployment, or verified product result.



Executive summary

The original university presentation proposed digital marketing for an AI-enabled traffic solution. This public edition converts that concept into a governed decision system. It defines the municipal buying committee, campaign taxonomy, qualification stages, privacy controls, AI claim controls, and an executable model for channel metrics and constrained budget recommendations.

Portfolio value

- Demonstrates B2G positioning beyond consumer-style impressions and clicks.
- Shows reproducible analytics with transparent assumptions and synthetic data.
- Connects marketing decisions to procurement readiness, privacy, security, and AI accountability.
- Separates promotional claims from operational traffic evidence.

Decision rule

Use channel scores to structure review, not to automate spending. A human owner must assess legal basis, data quality, buying-committee progress, channel saturation, and evidence status before reallocating budget.

1. Context and opportunity

European urban-mobility policy emphasises integrated planning, measurable indicators, sustainable mobility, and digital solutions. The commercial opportunity is therefore not a generic promise to use AI. It is a bounded pilot that helps a public buyer test a specific operational outcome with safety, governance, and procurement evidence.

Market signal	Implication for marketing
Sustainable Urban Mobility Plans	Align content with approved local mobility objectives and indicators.
Innovation procurement	Offer staged proof, acceptance criteria, and a credible route from pilot to scale.
AI governance	Explain intended purpose, oversight, evidence, and failure handling.
Data protection	Minimise campaign data and keep it separate from operational traffic telemetry.

Positioning

The recommended message is: test whether adaptive traffic management can improve agreed local outcomes inside a controlled pilot, with explicit baselines, human override, transparent evaluation, and documented conditions for scale.

2. Municipal buying committee

Role	Decision concern	Required evidence
Executive sponsor	Public value and accountability	Outcome baseline and citizen-impact narrative
Mobility owner	Safe and reliable operation	Pilot boundary, monitoring, fallback, operational KPIs
IT and security	Integration and resilience	Architecture, logs, access controls, recovery
Procurement	Fair competition and value	Scope, acceptance criteria, exit and ownership terms
DPO and legal	Lawful and proportionate processing	Data map, retention, DPIA trigger, contracts
Finance	Affordability and uncertainty	Pilot cost, TCO range, sensitivity and risk

A contact becomes a qualified opportunity only when an operational problem, accountable owner, plausible procurement route, and access to the necessary reviewers have been recorded.

3. Funnel and channel design

Stage	Buyer action	Marketing evidence	Exit criterion
Discover	Recognises a mobility problem	Outcome briefing and local context	Relevant stakeholder engages
Explore	Compares intervention options	Architecture and governance overview	Problem and owner confirmed
Qualify	Tests organisational fit	Readiness checklist and discovery workshop	Qualified lead criteria met
Evaluate	Builds internal case	Pilot plan, acceptance measures, risk register	Opportunity committee formed
Procure	Selects compliant route	Scope, pricing model, evidence pack	Procurement path documented
Pilot	Runs bounded evaluation	Baseline, monitoring, and review plan	Pilot commitment recorded

Channel roles

- LinkedIn: reach named professional audiences with outcome and governance content.
- Search: capture active intent around smart traffic pilots and procurement questions.
- Email: support account-based outreach with documented suppression and purpose rules.
- Webinars: bring the full buying committee into a technical evidence session.
- Innovation platforms: establish procurement and funding relevance.
- Trade media: build category awareness while controlling claims and attribution limits.

4. Measurement model

KPI	Formula	Use
CTR	clicks / impressions	Message-channel response
Engagement rate	engaged sessions / landing sessions	Content relevance proxy
Lead conversion	qualified leads / landing sessions	Qualification effectiveness
Opportunity conversion	opportunities / qualified leads	Buying-committee progression
Pilot conversion	pilot commitments / opportunities	Procurement and commercial progress
Cost per opportunity	spend / opportunities	Pipeline efficiency

UTM parameters provide campaign labels. They do not establish consent, identity, or causation. Contact-level activity remains in controlled systems. The public dataset contains aggregate synthetic counts only.

5. Synthetic scenario results

Channel	Engaged	CPL	CPO	Score	Budget share
Email	74.7%	€94	€282	91.0	22.3%
Webinars	79.8%	€169	€490	82.3	20.2%
Search	64.9%	€203	€900	70.1	17.2%
LinkedIn	60.6%	€270	€912	67.2	13.9%
Innovation platforms	67.6%	€252	€709	66.9	16.4%
Trade media	46.9%	€474	€1800	30.5	10.0%

Email and webinar activity perform strongly in the scenario because they create high-intent interactions with named buying committees. Search contributes greater volume. Trade media remains useful for awareness, but its example pipeline efficiency is weaker. These findings are generated from synthetic inputs.

6. Recommendation logic

Component	Weight	Rationale
Engagement	20%	Signals content relevance
Lead efficiency	25%	Rewards efficient qualification
Opportunity efficiency	25%	Rewards buying-committee progression
Strategic fit	20%	Preserves B2G and pilot relevance
Data confidence	10%	Reduces confidence in incomplete channels

Each channel receives at least 10% and at most 30% of the illustrative budget. This avoids abrupt concentration from a small sample. The allocation is an auditable heuristic. It does not estimate causal incremental return.

Required human review

- Confirm the campaign has a lawful and documented operating basis.
- Check whether conversions represent one account or many independent accounts.
- Review procurement timing and buying-committee coverage.
- Verify that product claims remain within owned evidence.
- Record the decision, owner, rationale, and review date.

7. Privacy and AI governance

Domain	Control	Evidence
Consent	Gate optional tags and preserve preference state	Configuration and test log
Minimisation	Collect campaign and funnel fields required for the stated purpose	Data dictionary
Retention	Delete or aggregate data on an approved schedule	Retention rule and deletion evidence
Claims	Link each capability or impact statement to an evidence owner	Claims register
AI oversight	Define intended purpose, intervention, logging, and fallback	System card and test plan
Separation	Keep marketing analytics outside operational traffic data flows	Architecture decision record

Classification under the EU AI Act depends on the intended purpose and actual deployment. A real supplier must obtain qualified legal and technical analysis rather than infer classification from marketing language.

8. Risk register

ID	Risk	Likelihood	Impact	Primary control	Owner
R1	Campaign tracking begins before valid consent	Medium	High	Consent management, tag gating, regional validation	Marketing Operations
R2	Public-sector contacts are used without a lawful outreach basis	Medium	High	Purpose review, suppression list, documented outreach rules	DPO and Legal
R3	Synthetic performance is presented as production evidence	Medium	High	Prominent labels, provenance notes, no customer claims	Portfolio Owner
R4	AI capability claims exceed tested system boundaries	High	High	Claims register, technical review, evidence owner	Product and AI Lead
R5	A channel is scaled from low-volume or biased data	Medium	Medium	Confidence threshold, minimum sample size, human review	Growth Lead
R6	Procurement journey is reduced to a consumer funnel	High	Medium	Buying-committee stages, tender readiness, long-cycle attribution	B2G Sales Lead
R7	Traffic-system deployment creates safety or resilience risk	Low	Very High	Pilot sandbox, fail-safe mode, simulation, rollback	Engineering and Municipality
R8	Personal data leaks through campaign exports	Low	High	Aggregate repository data, access controls, retention schedule	Data Owner

9. Ninety-day implementation roadmap

Period	Work	Exit evidence
Days 1–15	Confirm buyer problem, data boundary, claims status, and funnel definitions	Approved scope and data dictionary
Days 16–30	Build governed campaign registry, landing pages, and consent tests	Launch checklist and test log
Days 31–45	Publish evidence assets and run initial account-based outreach	Campaign registry and engagement baseline
Days 46–60	Run webinar and pilot-readiness discovery	Qualified account notes and objections
Days 61–75	Review channel quality, procurement stage, and claims	Decision record and experiment backlog
Days 76–90	Present pilot business case and controlled next-quarter allocation	Pilot proposal and approved budget

10. Limitations

- Campaign values are synthetic and are not market benchmarks.
- The scoring weights reflect a portfolio scenario and require sensitivity analysis in practice.
- Last-known campaign attribution does not establish causal contribution.
- Public-sector sales cycles may span budgets, elections, tenders, grants, and policy changes.
- Traffic-control safety, security, and AI performance are outside this marketing analysis.
- Jurisdiction-specific procurement and direct-marketing requirements are not legal advice.

Conclusion

A credible digital-marketing system for public-sector AI links every message to evidence, every campaign to a governed purpose, every lead to a buying committee, and every budget recommendation to accountable human review. The portfolio value lies in this operating model and its reproducible artefacts.

References

1. **European Commission. Sustainable urban mobility planning and monitoring. 2026.** https://transport.ec.europa.eu/transport-themes/urban-transport/sustainable-urban-mobility-planning-and-monitoring_en
2. **European Commission. The New EU Urban Mobility Framework. 2021.** https://transport.ec.europa.eu/system/files/2021-12/com_2021_811_the-new-eu-urban-mobility.pdf
3. **European Commission. Intelligent transport systems for roads. 2026.** https://transport.ec.europa.eu/transport-themes/smart-mobility/road_en
4. **European Commission. Innovation procurement. 2026.** https://commission.europa.eu/funding-and-tenders/tools-public-buyers/innovation-procurement_en
5. **European Union. Regulation (EU) 2024/1689, Artificial Intelligence Act. 2024.** https://eur-lex.europa.eu/eli/reg/2024/1689/oj
6. **European Union. Regulation (EU) 2016/679, General Data Protection Regulation. 2016.** https://eur-lex.europa.eu/eli/reg/2016/679/oj
7. **European Data Protection Board. Guidelines 05/2020 on consent under Regulation 2016/679. 2020.** https://www.edpb.europa.eu/sites/default/files/files/file1/edpb_guidelines_202005_consent_en.pdf
8. **Google Analytics Help. Collect campaign data with custom URLs. 2026.** https://support.google.com/analytics/answer/10917952
9. **NIST. Artificial Intelligence Risk Management Framework 1.0. 2023.** https://doi.org/10.6028/NIST.AI.100-1